Building a Facilitator Agent to Promote Understanding in Multilingual Communication

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01 Introduction

◆SDGs「Quality Education」

- To develop global citizenship, it is important to appreciate cultural diversity.

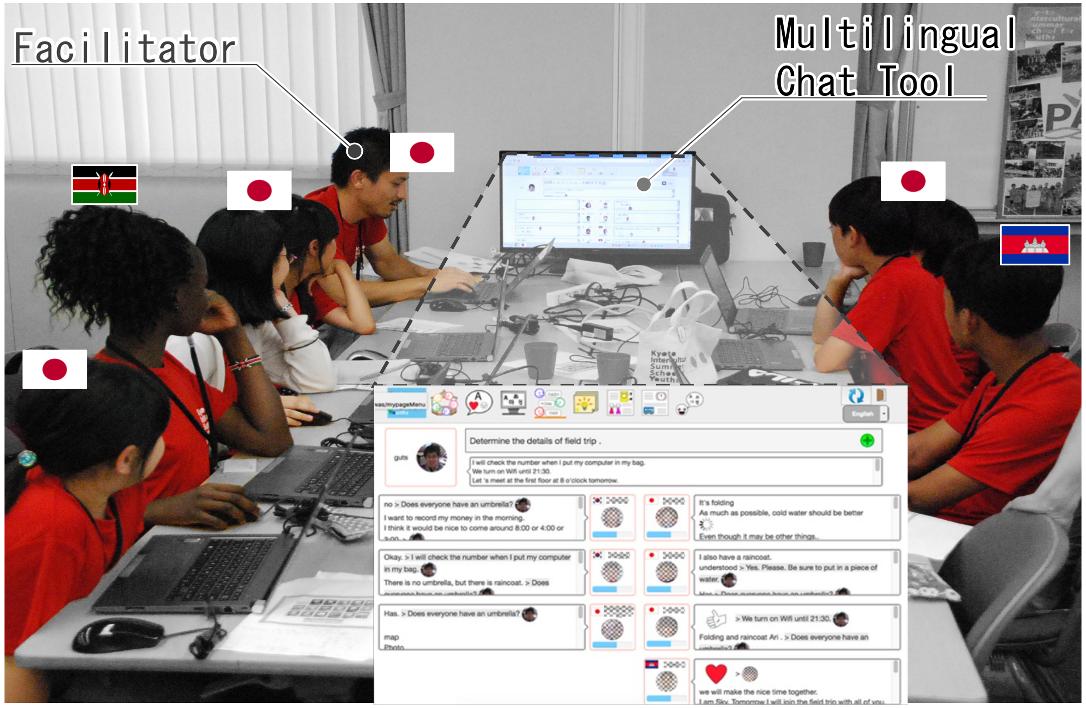
◆Children experience intercultural collaboration using machine translation

- To understand the others
- To make themselves understood

01 Introduction

◆Children's Intercultural Collaboration

- KISSY(Kyoto International Summer School for Youth)



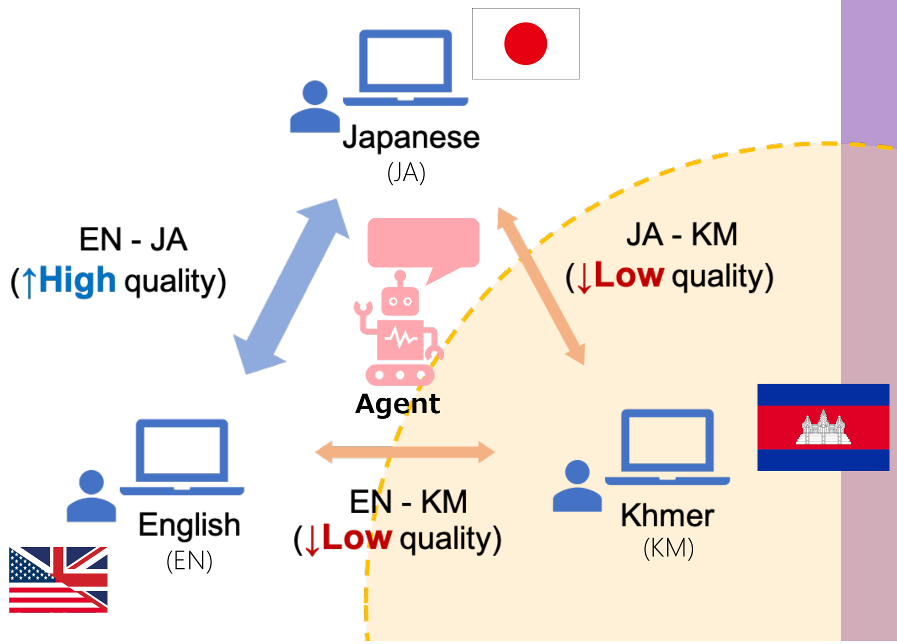
Source: PANGAEA

02 Purpose

◆ Children speaking low-resource languages(LRL) tend to speak less[1]

- **low-resource languages**: Languages with fewer language resources available for machine translation and lower translation accuracy.

◆ Building the two types of facilitator-agent to facilitate their understanding of the discussion.



[1] Mondheera Pituxcoosuvarn, Toru Ishida, Naomi Yamashita, Toshiyuki Takasaki, Yumiko Mori (2018): Machine Translation Usage in a Children's Workshop: Proceesings of The Eighth International Conference on Collaboration Technologies (CollabTech' 18): pp.59-73

02 Facilitator Agents

◆Two types of facilitator agents

Summary Request

Ask high-resource language speakers to summarize the discussion.
Ex: "I can't keep up with the discussion, so please summarize what's going on. "

Summary Provision

Summarize the discussion using ChatGPT
Ex: "We are now discussing XX. There is an opinion that this item is important for XX reasons. "

➡ Both agents operate after four consecutive utterances by high-resource language speakers.

03 Hypotheses

H 1

Both types of agents improve the comprehension of discussions by LRL speakers.

H 2

Both types of agents increase the participation of LRL speakers in discussions.

H 3

The summary-requesting agent increases high-resource language speakers’ consideration for LRL speakers.

04 Experiment

◆The communication experiment using the facilitator agents.

Experimental Contents

Subject: 4 groups(2 Japanese, 1 Nepali(LRL))

Task: Survival Task
(The participants discuss and rank the most important items for survival in difficult situations.)

Conditions: (1)Summary request agent
(2)Summary provision agent
(3)Without agent

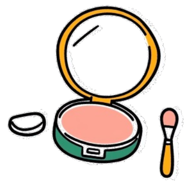
04 Survival Task

<Ex: Dessert Survival>

Scenario: You have crash-landed in a small airplane somewhere in the Arizona Desert. It is approximately 9:00 AM, in mid-August. The ground temperature will reach 54°C (130°F) degrees later today. You and two other members are the only survivors. The airplane may explode at any minute, so you had to evacuate as soon as possible. Before evacuating, you were able to salvage 3 items from the crash. All items are in good condition. Now, you must rank the 3 items in order of importance to your survival.

Items:
① .45 caliber pistol (loaded)
② A red and white parachute
③ A cosmetic mirror

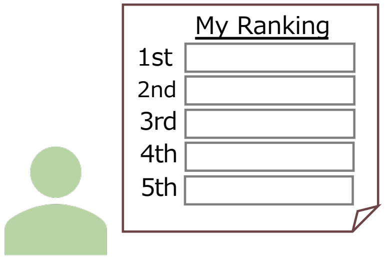





04 Experiment

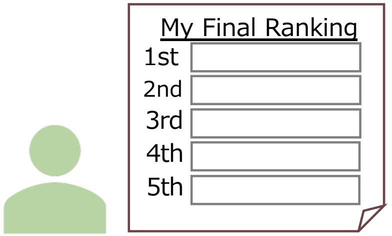
◆Each task consists of 4 steps.

Step 1 Each subject determines their **individual ranking**.



Step 2 The subjects discuss with other members and determine their **group ranking**.

Step 3 Each subject determines the final **individual ranking** based on the discussion.



Step 4 The subjects answer the questionnaire.

04 Experiment

◆The subjects communicate using a multilingual chat tool (Langrid Chat) with an embedded facilitator agent.

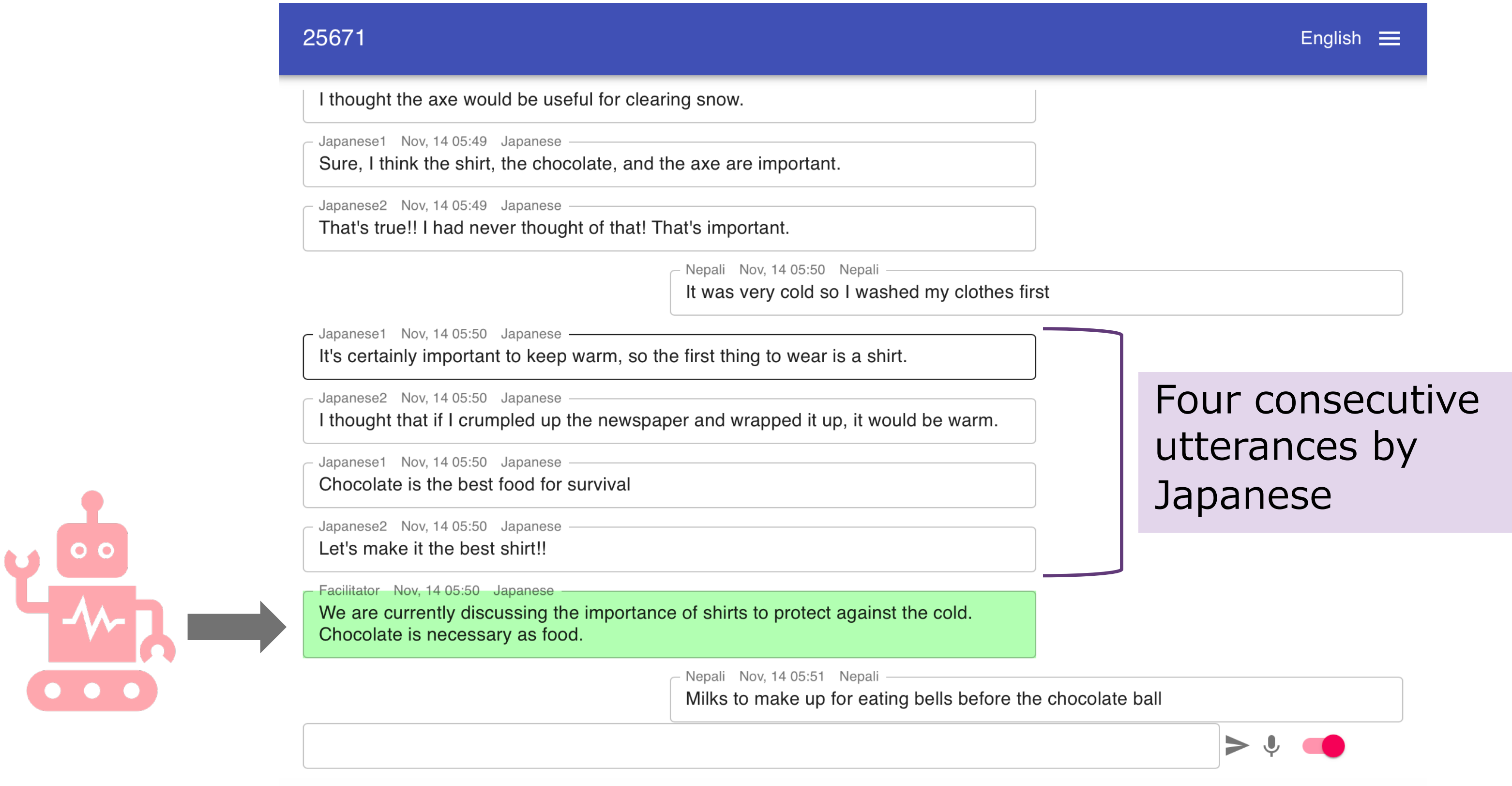


Figure: User Interface of a multilingual chat tool (Langrid Chat)

05 Analysis of Hypothesis 1

◆We analyzed the answers to each question about the discussion understanding of Nepali speakers.

Questions (on a 5-point scale) :

Q1. I perfectly understood the overall content of the discussion

Q2. I perfectly understood the messages from other participants.

Q3. Other participants perfectly understood my messages.

Q4. All participants completely understood each other’s thoughts.

Q5. The fact that participants had different native languages affected the understanding of the discussion.

05 Analysis of Hypothesis 1

◆A *U test* was conducted to compare data groups with and without agents.

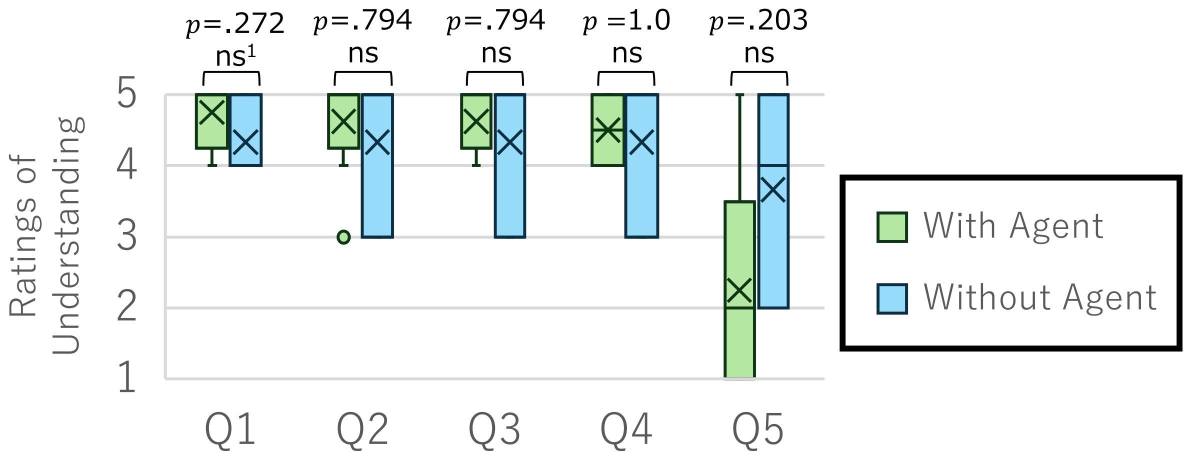


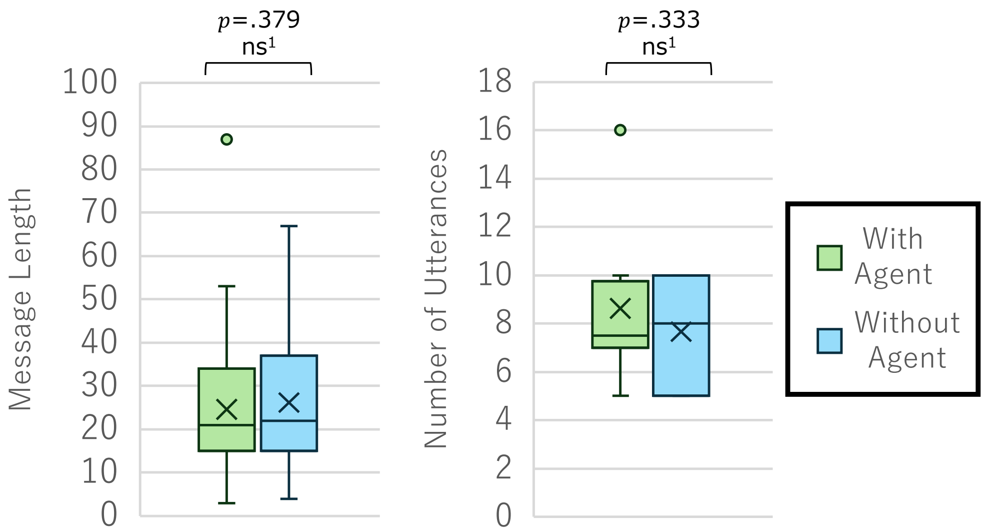
Figure: Results of the evaluation among conditions.

"*Q5. the fact that participants had different native languages affected the understanding of the discussion.*" tends to be rated higher without an agent.

06 Analysis of Hypothesis 2

◆The number and length of utterances during each task for Nepali speakers were analyzed.

◆A *U test* was conducted to compare data groups with and without agents.



(a) Message length. (b) number of utterances.

Figures: Results of the evaluation among conditions.

The number of utterances tends to be rated higher the with-agent condition.

06 Qualitative Analysis

◆We analyzed qualitatively based on utterances in the dialogue logs.

No.	Sender	Message
1	Nepali	You can put a coffee pot on the second one
2	Japanese 1	yes
3	Japanese 2	In second place, I think a flashlight would be food because light is necessary at night.
4	Japanese 2	I think a flashlight is an essential item, so I think it should be in second place.
5	Japanese 1	Since a coffee pot neither provides light nor drinks, I think number 4 is a good choice.
6	Facilitator	We are currently discussing the ranking of the items, with the flashlight being number two and the coffee pot being number four.
7	Japanese 2	I didn't think the coffee pot would get much use.
8	Nepali	We can use coffee pot to drink water
9	Japanese 1	I didn't think there was any water in the coffee pot.
10	Japanese 2	I don't think it's necessary since you can rescue it with your hands and drink the water.
11	Nepali	So let's put the light in the second place

After the facilitator provided a summary, he expressed his opinion about flashlight.

06 Qualitative Analysis

- We analyzed the ranking of Nepali individuals before the discussion and the group ranking using *Spearman's rank correlation coefficient*.
- A *t-test* was conducted to compare these data groups with and without agents

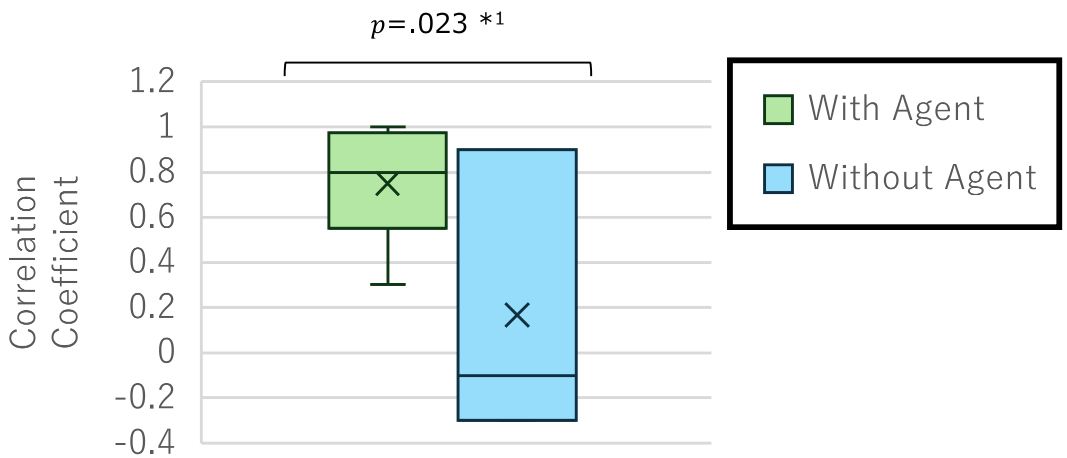


Figure: Results of the evaluation among conditions.

There is a significant difference. The agent enables comprehensive intervention in the discussion.

06 Analysis of Hypothesis 3

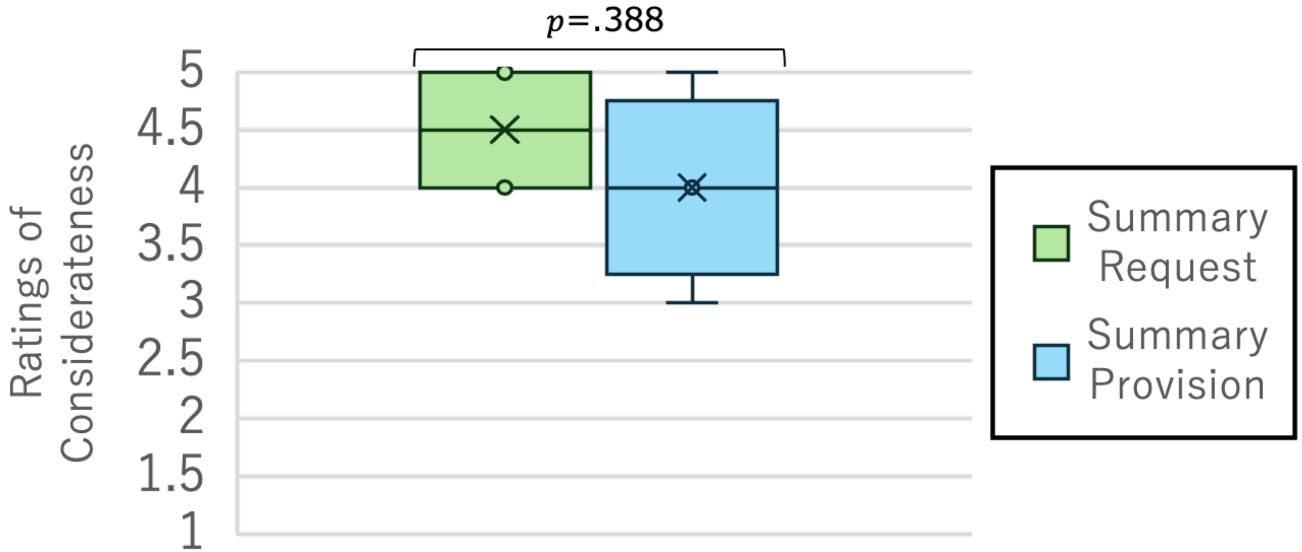
◆We analyzed the answers to each question about considering the LRL speaker of Japanese speakers.

Questions (on a 5-point scale) :

Q1. I took care to ensure that Nepali speakers could understand the content of the discussion.

06 Analysis of Hypothesis 3

◆A *U test* was conducted to compare data groups with and without agents.



Figures: Results of the evaluation among conditions.

“Q. I took care to ensure that Nepali speakers could understand the content of the discussion.” tends to be rated higher the summary request agent.

06 Conclusion

◆We proposed the agents with summary requests and summary provisions.

◆The effectiveness of these agents was verified through communication experiments with them.

Result

- In most statistical tests, there were no significant differences. However, observations of box-and-whisker diagrams and dialogues uncovered some trends.

- With these agents, the impact of different mother tongues on argument understanding was not perceived.
- LRL speakers had better comprehension with the agents.